

ENTRY 073

The True Zero-Shot Fourth-Chip Test — September 5, 2026 — Status: COMPLETE, HONEST MIXED RESULT

Osaka: A Chip the Model Has Never Touched

The Cleanest Cold-Transfer Test This Project Has Run

Why This Test Is Different From Every Prior One

Brisbane (Entry 061) was added specifically to test cross-chip generalization, but became a training chip almost immediately (Entry 065's combined model) -- so no result since has reflected a chip the DEPLOYED model has literally never seen. FakeOsaka (127 qubits, same heavy-hex class as the other three, 144 real calibrated edges) was set up and a 708-circuit eval-only bell-pair dataset was grown for it, but -- critically -- never folded into any training pipeline. The Entry 071 deployed model, trained on Kyiv+Sherbrooke+Brisbane, was scored cold against it. Osaka has no per-chip normalization statistics in the model at all; the average of the three training chips' stats was used as the best available substitute, since a real deployed tool facing an unfamiliar chip would need some answer here.

Results — Osaka, Zero Training Exposure

Metric	GNN (Entry 071)	v4.1 closed-form	GNN on trained chips (ref.)
MAE	5.82	5.57	1.38
R²	0.623	0.686	0.964
floor-collapse MAE (n=49)	15.77	22.38	0.50-0.56
mean predicted std	3.27 pts	n/a	0.81 pts

Two honest findings, not one. First: on aggregate accuracy, the GNN does NOT clearly beat the hand-derived v4.1 closed-form formula on a genuinely novel chip -- they are roughly tied, with v4.1 very slightly ahead on both MAE and R². This is worth stating plainly rather than downplaying: the GNN's advantage over the closed-form model, demonstrated repeatedly on chips it has trained on, does not automatically transfer to a chip with zero exposure. Second: on floor-collapse specifically -- the failure mode v4.1 structurally cannot see at all, and the entire reason the GNN was built (Entry 036 onward) -- the GNN still meaningfully outperforms v4.1 even zero-shot (15.77 vs 22.38 MAE), suggesting the floor-collapse detection capability generalizes better than raw point-accuracy does.

The Uncertainty Mechanism Passes Its Hardest Test

Mean predicted uncertainty on Osaka (3.27 pts) is roughly 4x wider than on the model's home chips (0.81 pts) -- and the actual MAE degradation is also roughly 4x (5.82 vs 1.38). The MC-Dropout uncertainty band, calibrated in Entry 071 on held-out data from the training chips, correctly and proportionally widens when the model is genuinely out of its depth on a chip it has never seen at all. This is the most important validation this uncertainty mechanism has had: it is not just calibrated on interpolation within known chips, it also degrades gracefully and honestly under true extrapolation, which is exactly the property a deployed tool needs to avoid confidently wrong answers on unfamiliar hardware.

Entry 073 — Conclusion

The true zero-shot test gives a mixed, credible answer rather than a clean win: the GNN's raw accuracy advantage over the closed-form baseline does not fully survive a genuinely unseen chip, but its floor-collapse detection and its uncertainty calibration both do. For a project whose stated long-term goal is a deployable tool anyone can query on any chip, this is the single most useful result to have on record before productization begins: it tells a future user exactly what to expect (roughly v4.1-level point accuracy, better floor-collapse catching, and honestly wide error bars) on a chip the model was never trained on, rather than an optimistic claim that would not hold up.

Research Log — Index Addendum (Entry 073)

Files: `setup_osaka.py`, `entry073_grow_osaka.py`, `entry073_zero_shot_test.py`. Data:
`quantumbridge_data/entry073_osaka_bell_dataset.json` (708 eval-only circuits, never trained on),
`entry073_osaka_graph_dataset.json`, `entry073_zero_shot_results.json`.